

Exercise 1: Using Venn diagram

According to a television survey 32% of South African households have DSTV. Seventy-four percent of SA households with a television set have more than one set. Suppose 20% of all South African households have DSTV and also more than one television set. A South African household is selected at random:

Events:

A = household has DSTV

B = household has more than 1 TV

- (a) What is the probability that a household has DSTV or more than one television set?
{Hint: $P(A \cup B)$ }
- (b) What is the probability that a household has DSTV or more than one television set, but not both?
{Hint: $P(A \cup B) - P(A \cap B)$ }
- (c) What is the probability that a household does not have DSTV and not more than one television set? {Hint: $P(A' \cap B')$ }
- (d) Are the events a household has DSTV and a household has more than one television set mutually exclusive? {Hint: $P(A \cap B)$ }
- (e) Construct the contingency table with probabilities.

Exercise 2: Application of Bayes theorem

Small cars use less petrol, but they are not as safe as bigger cars. Small cars accounted for 18% of the vehicles on the road, but accidents involving small cars led to 11898 fatalities during the recent year. The probability of a fatal accident, knowing it involves a small car, is 0.128 and the probability of a fatal accident, knowing it does not involve a small car, is 0.05.

Events:

A = event that it is a small car

B = event that it is a fatal accident

- (a) Suppose you hear of an accident involving a fatality. What is the probability that a small car was involved?
- (b) Construct the contingency table with probabilities.